

SUPERMARKET SALES ANALYSIS

BUSINESS ANALYTICS PROJECT

USING MICROSOFT EXCEL

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PROJECT OVERVIEW

This project focuses on analyzing supermarket sales data using Microsoft Excel and statistical analysis techniques. The analysis examines sales performance, quantity sold, product lines, customer characteristics, payment methods, ratings, and gross income.

The project uses data analysis, PivotTables, correlation analysis, standard deviation, and data visualization to identify patterns and relationships within the dataset. An interactive dashboard was also developed to present the key findings in a clear and visually accessible format.

The objective is to transform raw supermarket transaction data into meaningful business insights that can support better understanding of sales performance and customer behavior.

BUSINESS OBJECTIVE

The main objective of this project is to analyze supermarket sales data and identify meaningful patterns in sales performance, product demand, customer behavior, and financial performance.

The analysis aims to:

- Identify the total quantity of items sold.
- Determine the best and lowest-performing product lines.
- Analyze sales performance across different cities and customer segments.
- Examine the relationship between Unit Price and Quantity.
- Analyze the relationship between Rating, Gross Income, and Quantity.
- Measure the variation in Quantity and Unit Price using standard deviation.
- Create a dashboard to present important business findings visually.

DATASET DESCRIPTION

Dataset: Supermarket Sales Dataset

Number of Records: 1,000

Number of Columns: 17

Time Period: 2019

The dataset contains transaction-level information from a supermarket, covering sales, products, customers, payments, ratings, and financial measures.

KEY VARIABLES

Category	Variables
Transaction	Invoice ID, Date, Time
Location	Branch, City
Customer	Customer Type, Gender
Product	Product Line, Unit Price, Quantity
Sales & Finance	Tax 5%, Total, COGS, Gross Margin Percentage, Gross Income
Customer Feedback	Rating
Payment	Payment Method

The dataset was used to analyze product performance, sales trends, customer characteristics, and relationships between important business variables.

DATA CLEANING AND PREPARATION

Before performing the analysis, the dataset was reviewed and prepared in Microsoft Excel to ensure that the data was suitable for analysis.

The following steps were performed:

- Created a separate Cleaned Data sheet while preserving the original raw data.
- Checked the dataset for duplicate records.
- Checked all columns for blank or missing values.
- Verified that Quantity, Unit Price, and Rating were stored as numeric values.
- Checked the Date column to ensure that the dates were correctly recognized by Excel.
- Reviewed categorical fields such as Product Line, City, Branch, Gender, Customer Type, and Payment for consistency.
- Verified numerical fields such as Total, Tax, COGS, Gross Income, and Gross Margin Percentage.
- Added a Year column using the Date field to support year-based analysis.

After these checks, the cleaned dataset was used for PivotTables, statistical calculations, correlation analysis, and dashboard creation.

ANALYSIS AND METHODOLOGY

The analysis was performed using Microsoft Excel. PivotTables and statistical functions were used to examine the supermarket data and answer the business questions.

The following methods were used:

- **PivotTables** to summarize sales and quantity across product lines, cities, and other categories.
- **SUM and AVERAGE functions** to calculate key sales and performance measures.
- **CORREL function** to measure the relationships between selected numerical variables.
- **STDEV.S function** to measure the variation in Quantity and Unit Price.
- **Data visualization** through charts to identify and communicate important patterns.
- **Dashboard development** to present the major findings in a clear and visually accessible format.

The analysis was conducted systematically from data preparation to statistical analysis, followed by visualization and interpretation of the results.

KEY RESULTS

Q1. How many items were sold in a year?

The total quantity of items sold in 2019 was 5,510 units.

Q2. Which product line has the maximum quantity sold in every city?

Mandalay: Sports and Travel — 322 units

Naypyitaw: Food and Beverages — 369 units

Yangon: Home and Lifestyle — 371 units

Q3. Which product line has the minimum quantity sold in a year?

Health and Beauty had the lowest quantity sold, with 854 units.

Q4. Correlation between Unit Price and Quantity

The correlation coefficient is 0.0108.

This indicates an almost negligible positive linear relationship between Unit Price and Quantity.

Q5. Sales report for every product line

Product Line	Quantity	Total Sales	Gross Income
Electronic Accessories	971	54,337.53	2,587.50
Fashion Accessories	902	54,305.90	2,586.00

Food and Beverages	952	56,144.84	2,673.56
Health and Beauty	854	49,193.74	2,342.56
Home and Lifestyle	911	53,861.91	2,564.85
Sports and Travel	920	55,122.83	2,624.90

Food and Beverages generated the highest total sales and gross income among the product lines.

Q6. Standard deviation of Quantity and Unit Price

Quantity: 2.9234

Unit Price: 26.4946

Unit Price has a considerably greater spread than Quantity.

Q7. Relationship between Quantity and Unit Price

The correlation coefficient is 0.0108, indicating that there is almost no linear relationship between Quantity and Unit Price in this dataset.

Q8. Relationship between Rating and Gross Income

The correlation coefficient is 0.0364.

This indicates an almost negligible positive linear relationship between customer Rating and Gross Income.

Q9. Relationship between Rating and Quantity

The correlation coefficient is -0.0158.

This indicates an almost negligible negative linear relationship between customer Rating and Quantity.

BUSINESS INSIGHTS

The analysis produced several important business insights:

- **Food and Beverages performed strongly**
Food and Beverages generated the highest total sales (**56,144.84**) and gross income (**2,673.56**) among the product lines.
- **Health and Beauty had the lowest quantity sold**
Health and Beauty recorded **854 units**, the lowest quantity among all product lines, indicating an opportunity to investigate customer demand and product performance.
- **Product performance differs across cities**
The highest-selling product line by quantity was different in each city. Sports

and Travel performed best in Mandalay, Food and Beverages in Naypyitaw, and Home and Lifestyle in Yangon.

- **Unit Price has little influence on Quantity**

The correlation between Unit Price and Quantity was only **0.0108**, suggesting that changes in unit price were not strongly associated with the quantity purchased in this dataset.

- **Customer Rating showed very weak relationships with financial and sales variables**

Rating had almost no linear relationship with Gross Income (**0.0364**) or Quantity (**-0.0158**). This suggests that higher customer ratings did not necessarily correspond to higher sales quantities or gross income.

- **Unit Price showed greater variation than Quantity**

The standard deviation of Unit Price (**26.49**) was much higher than that of Quantity (**2.92**), indicating greater variability in product pricing.

Overall Insight

The analysis shows that product line and city are useful dimensions for understanding supermarket performance, while the tested relationships between price, quantity, ratings, and gross income were generally very weak.

DASHBOARD

An interactive Supermarket Sales Analysis Dashboard was developed in Microsoft Excel to present the key findings of the analysis in a clear and visual format.

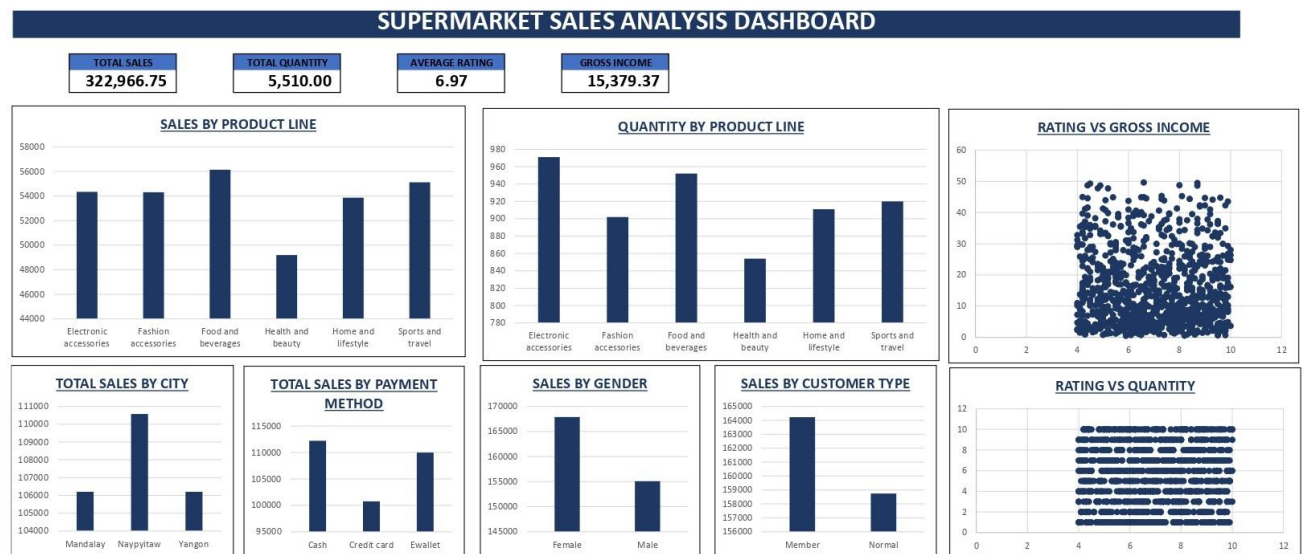
Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

KPI	Value
Total Sales	322,966.75
Total Quantity	5,510
Average Rating	6.97
Gross Income	15,379.37

Dashboard Visualizations

An Excel dashboard was developed to present the key findings of the analysis through KPIs and visualizations.



These visualizations allow users to quickly compare different aspects of supermarket performance and identify important patterns in the data.

RECOMMENDATIONS

Based on the analysis, the following recommendations can be considered:

- Focus on high-performing product lines**
 The business can continue to support strong-performing categories such as **Food and Beverages** while monitoring their sales and profitability.
- Investigate Health and Beauty performance**
 Since Health and Beauty recorded the lowest quantity sold, the business could examine customer preferences, product assortment, pricing, and promotional strategies for this category.
- Use city-level analysis for decision-making**
 Since different product lines performed best across different cities, inventory and promotional strategies can be adapted according to local demand.
- Monitor pricing independently from quantity**
 Since the correlation between Unit Price and Quantity was very weak, pricing decisions should consider other factors such as product type, customer preferences, competition, and promotions.
- Use customer ratings as a service-quality indicator**
 Although ratings showed very weak relationships with sales and quantity, they

can still be monitored to identify customer satisfaction issues and improve the overall shopping experience.

- **Use the dashboard for regular performance monitoring**

The Excel dashboard can help management quickly review sales, quantity, customer characteristics, payment methods, and other key performance indicators.

TOOLS AND SKILLS DEMONSTRATED

This project demonstrates practical application of the following Business Analytics skills:

Tools Used

Microsoft Excel

PivotTables

Excel formulas and statistical functions

Data visualization

Dashboard development

Skills Demonstrated

Data Cleaning and Preparation

Exploratory Data Analysis

Descriptive Statistics

Correlation Analysis

Standard Deviation Analysis

Business Data Interpretation

Data Visualization

Dashboard Design

Business Insight Generation

Analytical Problem Solving

This project demonstrates the ability to transform raw transaction data into structured analysis and communicate business findings through visualizations and a dashboard.

CONCLUSION

This project analyzed supermarket transaction data using **Microsoft Excel** and statistical analysis techniques. The analysis covered product performance, sales, quantity, customer characteristics, payment methods, ratings, and gross income.

Using **data cleaning, PivotTables, correlation analysis, standard deviation, and data visualization**, the project transformed raw transaction data into meaningful business insights. The analysis identified differences in product performance across cities and showed that the relationships between Unit Price, Quantity, Rating, and Gross Income were generally very weak.

The dashboard provides a concise visual summary of the major findings and demonstrates how Excel can be used to support **data-driven business analysis** and decision-making.

Overall, this project strengthened practical skills in Excel-based Business Analytics, statistical analysis, data visualization, and business insight generation.

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Project: Supermarket Sales Analysis

Tools: Microsoft Excel | Statistics | Data Visualization